

Martin Rodrigo Morales

AI Technical Lead · MLOps Engineer · AI Systems Architect

MSc Artificial Intelligence · Lic. en Psicología · B.Sc. Systems Engineering

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PROFESSIONAL SUMMARY

AI Technical Lead & MLOps Engineer with **5+ years** building production-grade LLM and agentic systems. Currently at **latel** leading a 4-person team on **+OAA / NOA**, an event-driven multi-agent platform running in production for a leading Spanish telecom operator, automating **root-cause** diagnosis of network failures. Co-founder and technical coordinator of **Dualtherapist** (clinical AI), approaching launch. **Architect** profile: agentic system design (MCP, A2A, BPM), RAG/GraphRAG, **LLMOps** (in-house gateway + self-hosted model), and end-to-end cloud architecture. Proven track record in **Fortune 500** environments (Accenture, BBVA/HP) and a unique position at the intersection of Systems Engineering and Clinical Psychology.

PROFESSIONAL EXPERIENCE

AI Technical Lead

Apr 2026 – Present

latel — Telecommunications / Enterprise AI · Spain

- Leading a 4-person team building **+OAA / NOA**, a production multi-agent platform that automates root-cause detection of network failures for a leading Spanish telecom operator.
- Event-driven microservices** architecture: multi-channel ingestion (syslog, mail, SMS, Telegram) → **Kafka (Avro)** + **Apache Flink** → Elasticsearch, with agent orchestration via **Flowable BPM (BPMN)** and an agent HTTP bridge.
- Designed the in-house **LLMOps** layer: **LiteLLM** + **Langfuse** gateway (governance, virtual keys, cost control and observability) and a self-hosted **Qwen3-30B model on GKE**, with a RAG vs. fine-tuning/distillation (RAFT) strategy.
- LLM reasoning pipeline over live network telemetry for root-cause classification and an **agentic chat with MCP** and multi-step reasoning. Stack: Next.js/React 19 (Nx), FastAPI + Postgres, OpenTelemetry, Docker, Kubernetes + Helm.

Independent AI Consultant & Professor

2025 – Present

Northern International University / Self-employed

- Appointed professor at NIU delivering modules on AI-Assisted Psychometrics, Computational Psychopathology, and Conversational Therapeutic Systems.
- Built scalable RAG pipelines for the health and education sectors integrating HuggingFace LLMs with Qdrant/ChromaDB vector stores.
- Mentored 20+ students through the full NLP project lifecycle: LangChain architecture, FastAPI deployment, and production best practices.

AI Engineer

2024 – 2025

Accenture — Client: HP (Fortune 500)

- Designed and deployed a RAG-based candidate evaluation system (MinHash near-dedup, perplexity filtering, domain rebalancing), improving **NDCG@5 from 0.71 to 0.83**.
- Led MLOps strategy for HP supply-chain predictive models: PySpark pipelines, Docker + Databricks CI/CD.
- Optimized Kubernetes cluster resource allocation and autoscaling, reducing cloud infrastructure costs.

Lead Data Instructor & Technical Mentor

2023 – 2025

Henry Bootcamp

- Led 10+ cohorts as Technical Lead & Scrum Master through 45-day production-simulation sprints.
- Co-designed cloud-native architectures (Docker, FastAPI, MLflow) deployed by graduates to real production environments.

SKILLS

LLM & AGENTS

LangChain LangGraph
Google ADK MCP A2A
RAG/GraphRAG LiteLLM Langfuse
HuggingFace Neo4j Qdrant

MLOPS & INFRA

Docker Kubernetes Helm
Terraform MLflow CI/CD
FastAPI OpenTelemetry Redis

DATA & STREAMING

Kafka Apache Flink Elasticsearch
Postgres PySpark Databricks
SQL

CLOUD

GCP Vertex AI Gemini Firebase
AWS Azure ML

DEV STACK

Python Next.js React Native
Expo/Solito Turborepo Nx Git

EDUCATION

MSc Artificial Intelligence

Universidad Tecnológica Nacional

AI & ML

B.Sc. Systems Engineering

Universidad Tecnológica Nacional

Engineering

Lic. en Psicología

Universidad de Mendoza — 2017

🏆 Best Graduate

CERTIFICATIONS

- Microsoft Azure AI (AI-900)
- IELTS C1
- HuggingFace Agents Course
- Cognitive Neuroscience (Neuro-Transmitiendo)

LANGUAGES

English C1 — IELTS
Spanish Native
French A2

Big Data Engineer

2023 - 2024

Accenture — Client: BBVA (Banking)

- Orchestrated mission-critical ETL pipelines in a regulated banking environment using Hadoop, Spark on YARN, and GCP Dataproc.
- Diagnosed and fixed dataloader starvation (repartition to 256 MB files + speculative execution), improving throughput ~2.5x; Data Quality & Drift monitoring with Control-M.

SELECTED PROJECTS

Dualtherapist — Clinical AI Platform

Co-founder & Tech Coordinator

Turbo monorepo · Next.js 15 · Expo/Solito · FastAPI · Gemini/Vertex AI · Neo4j · Firestore · Docker/K8s/Terraform

- Coordinating a 4-person team (frontend, backend, PM) building a digital therapy platform connecting psychologists and patients, approaching launch; responsible for technical architecture and agent system design.
- End-to-end architecture: Next.js 15 web (App Router/SSR) + Expo mobile app with universal components (Solito), a Node/Express API, and a Python/FastAPI ai-service for session transcription, embedding-similarity triage, and a RAG knowledge base powered by Gemini/Vertex AI.
- Clinical knowledge graph in Neo4j; Firestore + Google Cloud Storage; Firebase Auth + JWT HttpOnly cookie with Redis blacklist; Docker / Kubernetes / Terraform infrastructure.
- Designed the HITL evaluation and the domain-specific clinical safety layer, drawing on my training as a licensed psychologist.

Multi-Agent Knowledge Graph Builder

Google ADK 1.5 · LiteLLM · Neo4j · Python 3.11

- Orchestrated autonomous agents for entity extraction and Neo4j knowledge-graph construction from unstructured data.
- Two-stage chunking (coarse semantic + fine structural) improved entity extraction by ~8 F1 points; full pipeline from document ingestion to graph queries.

MetaAgent — Hybrid Memory AI System

LangChain · FAISS + Qdrant · OpenAI API · Docker

- Persistent AI assistant with hybrid vector memory (FAISS + Qdrant), human-feedback scoring loops, and long-term context retention.
- Containerized with Docker; session persistence with automated memory consolidation and a scalable deployment pipeline.

Production MLOps Pipeline — Sentiment Analysis

DistilBERT · FastAPI · MLflow · HuggingFace Spaces · CI/CD

- Deployed a production DistilBERT model (80% accuracy) on HuggingFace Spaces with a Gradio UI and automated CI/CD for model versioning.
- Training optimization (fp16 + gradient checkpointing: memory 38→22 GB, 4h→2.6h) and INT8 quantization with ONNX Runtime (500→130 MB, p99 CPU latency 280→95 ms).

PREVIOUS EXPERIENCE

Junior Data Scientist

2021 - 2023

CEA Solidario (NGO)

- Built AI-driven behavioral analysis tools for autism assistance, integrating early ML models for pattern recognition and behavioral insights.

CONTINUOUS DEVELOPMENT

- AI techniques: Prompt Engineering, Fine-tuning, RAG, Transfer Learning — HuggingFace Agents Course.
- React Native (Expo), advanced backend scalability, AWS serverless architecture (VPC, IaC).
- Specialization in Cognitive Neuroscience — Neuro-Transmitiendo (2024).

CLINICAL DOMAIN EXPERTISE

Licensed Psychologist with 5+ years of clinical experience. Applies diagnostic precision to LLM evaluation (HITL), hallucination detection, and AI safety design for mental-health and therapeutic AI systems.

RESEARCH BACKGROUND

APOLA

Structural Psychoanalysis — International Forums.

Topology

Set Theory, Logic & Language formalization.

KG Theory

Topological structures informing Knowledge Graph design.

Speaker

Research papers at international conferences.

DOMAIN FOCUS

- Mental Health AI
- Clinical NLP
- AI Alignment
- HITL Evaluation
- HealthTech
- EdTech
- FinTech
- Telco / Networks

LEADERSHIP & SOFT SKILLS

- Technical Lead & Scrum Master
- Cross-domain thinking: Tech × Clinical
- Fast learner — top academic track record
- Mentorship & knowledge transfer

🏆 ACADEMIC EXCELLENCE
Gold Medal — Best Graduate
Universidad de Mendoza, 2017